

NCRTC Bus Management System

Evaluated against the **NCRTC BMS Student PRD (Lite v1.0)** · Repository github.com/ItsAryanJ/intelligent-fleet-management-system

Evaluated **7 July 2026** · HEAD **d36ea82** (6-Jul) · Method: 16-agent adversarial code review with DB→API→UI wiring verification

OVERALL SCORE V3

7.6_{/10}

vs v2 6.7 → +0.9

Trajectory 4.4 → 6.7 → 7.6

Seams stitched, publish snapped

All three of v2's highest-leverage fixes landed and are genuinely wired end-to-end — the AVLS history/replay page, the week-roster grid with a live Publish, and geo-tagged panic feeding a full incident lifecycle. What holds v3 back is narrower and sharper: one high-visibility dead-end in CMS, a still-orphaned must-have, and no tests or CI at HEAD.

HOW THIS WAS SCORED

Sixteen agents read the current source — not the README's claims — each tracing whether a requirement is genuinely wired end-to-end (database model → API route → frontend call → rendered component with a real handler) versus stubbed, orphaned, decorative, or missing. A regression sweep re-checked every v2 residual gap; adversarial verifiers were told to *refute* each module's headline claim by default. The PRD's exact rubric weights were then applied and any requirement flagged fake was discounted rather than credited from the README. Findings were reconciled against independent manual spot-checks of the load-bearing claims. Graded as an MVP-depth student capstone, with fair partial credit for honest partials.

01 RUBRIC BREAKDOWN

CATEGORY	WEIGHT	V2	V3	RATIONALE
Functional completeness	40%	6.5	7.1	Four of five modules now trace end-to-end. AVLS (live WS + real replay), IMS (6-state workflow + geo-panic + assign) and Scheduling (week grid + wired Publish) genuinely closed their v2 gaps; auth + depot scoping stay broad and real. CMS drags: targeting is fixed, but create→publish is a functional dead-end and the must-have read-receipt roster is orphaned. Route ordered-stops/arrival editor and route delete are still absent.
Code quality & structure	15%	6.5	6.8	Clean feature-sliced routers, a typed exception hierarchy behind one structured handler, and a genuinely good GPS simulator (haversine, deadlock retry, deterministic lock order). Offset by ~2× scope bloat (20 tables / 18 routers vs the Lite PRD's 11), <code>config.py</code> printing the DB URL incl. password on every load, an unguarded weak default <code>SECRET_KEY</code> , dead RBAC permission tables, and an orphaned geofence router.
UI / UX & mobile	15%	6.5	7.2	Real ref-based Leaflet maps, a functional replay scrubber with speed controls, a genuine 7×4 week grid, and an incident drawer with status-gated buttons and a required-notes field. The wiring is real, not decorative. Docked below 8 for a thin live side-panel (no driver/route/trailing polyline), client-only status filters, and no deliberate mobile/responsive work on a driver-facing product.
Documentation & deliverables	10%	6.5	6.8	<code>ARCHITECTURE.md</code> with 9 Mermaid diagrams, plus API / DATABASE / DOCKER docs and wired Swagger + ReDoc; README credentials match the seed exactly. Penalized for a fabricated <code>backend/tests/</code> directory in the README, a broken Quick Start (<code>cd test1</code> into a nonexistent dir), a vestigial Alembic dependency with no migrations, and zero screenshots or demo video.
Demo & presentation	15%	6.0	7.0	<i>Scored code-only (no live demo/video available).</i> A fresh clone would largely run: docker-compose is genuinely health-gated (db → backend → frontend via <code>service_healthy</code>), the seed is idempotent, nginx proxies API + WS, and every scripted demo route exists. Hit by the broken README <code>cd</code> step (manual fix needed), a Dockerfile seed that swallows errors, and the copilot presented as live AI.
Honesty & judgment	5%	5.0	6.0	The headline v2 dishonesty is genuinely fixed: <code>/system/health</code> now runs real DB / WS / simulator / psutil probes, the redis fabrication is gone, and the copilot is honestly typed <code>mode='demo'</code> server-side. Residual: that caveat is captured into dead <code>copilotMode</code> state and never rendered while the UI markets "Fleet AI Copilot / Online"; Alembic is phantom; the README fabricates a <code>tests/</code> dir and the added CI <code>test.yml</code> was deleted in the final commit.

Weighted total: $(7.1 \times .40) + (6.8 \times .15) + (7.2 \times .15) + (6.8 \times .10) + (7.0 \times .15) + (6.0 \times .05) \approx 6.97$
base, **+0.65 net stretch credit** (WebSocket + trip replay full, SLA partial) = **7.6 / 10**. Demo assumption: no live demo or video available; the category was judged purely on whether a fresh clone would run the scripted PRD demo flow.

02 PER-MODULE SCORES

MODULE (PRD)	V2	V3	END-TO-END ASSESSMENT
Auth & Roles + depot scoping	8	7.5	Yes
Still the foundation. Real bcrypt, DB-backed login, working access + refresh with a genuine 401 frontend interceptor, consistent server-side depot scoping across ~13 routers. The <code>GET /routes</code> leak is only partly closed (still exposed to driver/conductor). Persistent gaps drop it slightly: dead RBAC permission tables (static map baked into the JWT), no per-request DB re-validation, unguarded weak default <code>SECRET_KEY</code> , no-op backend logout.			
AVLS Live Map & History	6	8	Yes
The biggest jump — v2 gap (a) fully closed. <code>HistoryPage.tsx</code> is real, routed at <code>/avls/history</code> , nav-linked, and calls the previously-orphaned <code>/gps/history</code> + <code>/trip-analytics</code> ; it draws segmented day polylines with a working play / pause / scrub + 1x–10x replay. The live map is real WebSocket animation end-to-end. Shortfalls: depot filter is client-side only, and the marker side-panel omits driver / route / trailing polyline.			
IMS Incidents	6	8	Yes
v2 gaps (c) and (d) fixed. Panic hits the dedicated <code>/incidents/panic</code> with real <code>navigator.geolocation</code> and a graceful fallback; a full 6-state machine (OPEN→ACK→ASSIGNED/IN PROGRESS→RESOLVED→CLOSED) is enforced by server-side <code>VALID_TRANSITIONS</code> ; manager assign is wired with a permission gate; resolve enforces notes. Real SLA compute + dashboard. Residual: photo upload is absent (<code>FileUpload</code> has zero importers) and notes are enforced only on resolve.			
Scheduling Routes & Duties	5.5	7	Partial
v2 gap (b) fixed: a genuine 7-day × 4-shift week grid with navigation; the Publish button invokes <code>POST /duties/publish</code> (flips DRAFT→PUBLISHED and emits notifications) — <code>publishMutation</code> is no longer dead. Acknowledge writes a real timestamp. Residual: the purpose-built <code>GET /duties/roster</code> is itself orphaned (the page uses <code>/duties</code>); no ordered-stops / arrival editor; no route delete; double-booking is detected post-hoc, not prevented.			
CMS Notices	5	5.5	Partial
Targeting is genuinely fixed end-to-end for ALL / ROLE / DEPOT (form → <code>POST</code> → <code>_can_view_notice</code> feed filter). But two blockers keep it near flat: (1) create→publish is a dead-end — new notices default <code>is_published=False</code> , yet the admin page fetches <code>/notices/feed</code> which returns only published rows, so the draft never appears and its Publish button can never render; (2) the must-have read-receipt roster (<code>/notices/{id}/readers</code>) is orphaned — the UI shows only an aggregate count.			

▲ Verified working end-to-end

- **Live AVLS map** with real WebSocket-driven marker animation — JWT-authenticated WS, backoff reconnect, rAF `slideTo`.
- **GPS history replay**: segmented day polylines, start/end markers, play/pause/skip/scrub, 1x–10x speed, trip analytics.
- **IMS full lifecycle**: geo-tagged panic, ack/assign/in-progress/resolve/close with transition guards, required resolve notes, live SLA dashboard.
- **Scheduling week grid** with navigation, wired Publish (DRAFT→PUBLISHED + notifications), real `acknowledged_at` timestamps.
- **Auth**: real bcrypt, DB-backed login, access+refresh with a working 401 retry interceptor, broad server-side depot scoping.
- **Health-gated compose**, idempotent seed, README credentials matching the seed, an honest real-probe `/system/health`.

▼ What remains (holds it off 8+)

- **CMS publish dead-end** — created notices default unpublished but the feed returns only published, so the Publish button never renders and drafts vanish from the UI.
- **Read-receipt roster** (a PRD must-have) still absent — `/notices/{id}/readers` orphaned; only an aggregate count shows.
- **No tests, no CI** — the added `test.yml` and `test_e2e.py` were deleted in the final commits; the README still fabricates a `backend/tests/` dir.
- **Phantom Alembic** — dependency only; schema still built at runtime via `create_all`.
- **Copilot honesty** — demo caveat never reaches the UI (dead state) while marketed as live "Fleet AI Copilot / Online".
- **Scheduling depth** — no ordered-stops / planned-arrival editor, no route delete; double-booking detected post-hoc.
- **Orphaned bloat** — geofence router (4 endpoints), `GET /duties/roster`, incident photo upload — all with zero callers.

04 V2 → V3 PROGRESS & REGRESSION SWEEP

Each known v2 residual gap, re-traced DB→API→frontend against the current source. **No regressions detected.**

GAP	V2 RESIDUAL DEFECT	V3 STATUS	EVIDENCE
a	AVLS history/replay page entirely absent on the frontend (orphaned <code>/gps/history</code> , <code>/trip-analytics</code>)	Fixed	<code>HistoryPage.tsx</code> routed at <code>App.tsx:64</code> , nav-linked, calls both endpoints, renders replay scrubber + segmented polylines.
b	Roster a single-date card list, not a weekly grid; Publish dead (<code>publishMutation</code> never invoked, <code>POST /publish</code> orphaned)	Fixed	<code>RosterPage.tsx</code> is a 7-day × 4-shift grid; Publish invokes <code>POST /duties/publish</code> → flips status + emits notifications.
c	Panic posted the generic <code>/incidents</code> and captured no location	Fixed	<code>DriverDashboard.tsx</code> → <code>POST /incidents/panic</code> with <code>navigator.geolocation</code> ; backend <code>router.py:281</code> dedicated endpoint.
d	Incident status workflow collapsed to Open→Resolved; manager assign orphaned; notes not enforced	Fixed	Server-side <code>VALID_TRANSITIONS</code> 6-state machine; assign wired with permission gate; resolve enforces notes.
e	CMS could only target ALL; role/depot targeting broken; read-receipt roster (must-have) absent	Partial	Targeting fixed E2E via <code>_can_view_notice</code> . But the publish path dead-ends and the read-receipt roster is still orphaned.
f	<code>/system/health</code> hardcodes <code>redis/websocket</code> "active"; copilot 200-on-error; "demo mode" caveat never reaches UI	Partial	Health now runs real probes; redis fabrication removed; copilot honestly typed server-side. But the demo caveat still never renders in the UI.
g	Shallow smoke tests that skip; no CI	Still broken	<code>test.yml</code> + <code>test_e2e.py</code> added then deleted in the final commits; no <code>.github/workflows</code> at HEAD; README still fabricates <code>tests/</code> .
h	Alembic migrations referenced but absent	Still broken	Dependency present but no <code>versions/</code> ; schema still built via <code>create_all</code> at runtime.
i	<code>docker compose up</code> seeding reliability	Fixed	Existence-guarded idempotent seed; compose health-gated across db → backend → frontend.

5 Fixed

2 Partial

2 Still broken

0 Regressions

05 STRETCH CREDIT

STRETCH ITEM	AVAILABLE	V2	V3	REASON
WebSocket live updates	+3%	Earned	Earned	Real end-to-end: simulator broadcast → WS <code>/gps/stream</code> (4001 auth) → <code>useGPSSocket</code> CustomEvent → rAF <code>slideTo</code> marker animation. Full credit.
Geofence breach detection	+3%	Denied	Denied	The geofence / route-deviation router (4 endpoints) has zero frontend callers — orphaned bloat, no breach banner. No credit.
SLA timers + breach alerts	+2%	Earned	Partial	Real per-severity deadline compute and a live breached / at-risk / on-track dashboard. Docked (+1.5) because the <code>sla_breached</code> badge only persists at resolve time.
Trip replay scrubber	+2%	Denied	Earned	Now a genuine wired replay: segmented polylines, start/end markers, a working play/pause/scrub with 1x–10x speed. Full credit.

Net stretch credit $\approx +6.5$ of 10 possible → contributes $\sim +0.65$ to the /10 score. WebSocket and trip replay are full; SLA partial; geofence denied.

06 HIGHEST-LEVERAGE FIXES (TO REACH 8+)

- Fix the CMS publish path** — set `is_published` on create, or point the admin list at `GET /notices` (which returns drafts) so the create→publish flow and the Publish button become reachable end-to-end. The cheapest single win.
- Wire the read-receipt roster** — call `GET /notices/{id}/readers` in `NoticeDetailDrawer` to render who read/acknowledged vs still-unread. Closes the last must-have.
- Restore real tests + CI** — re-add `test.yml` and a genuine backend suite with assertions; remove the fabricated `tests/` claim from the README.
- Surface the copilot demo caveat** in the UI (render `copilotMode` as a badge) and stop labeling it "Online AI".
- Close the route leak** — add a `ROUTE_VIEW` gate + universal depot filter so driver/conductor stop receiving all depots' routes; guard the default `SECRET_KEY` at startup.
- Add the ordered-stops + planned-arrival editor** for routes (schema exists, no endpoint/UI) to close the last core Scheduling gap; fix the README `cd test1` target.

FAIRNESS NOTE

Graded as a student capstone against the deliberately scoped-down, MVP-depth Lite PRD — not a production system. Rubric weights were applied exactly as specified (Functional 40 / Code 15 / UI 15 / Docs 10 / Demo 15 / Honesty 5). Functional completeness was weighted heaviest; honest partial implementations received fair partial credit, and any requirement the adversarial verifier flagged as stubbed, orphaned, or fabricated was discounted rather than credited from the README. Demo & presentation was scored from code-only demoability evidence — that assumption is stated and is the primary residual constraint on that category. v2 was graded at HEAD 33b1231 (26-Jun); v3 reflects HEAD d36ea82 (6-Jul), five feature commits later.